SIOC 217C: Climate and Climate Change

Homework #3

Consider an atmosphere filled by an absorber with uniform mixing ratio that absorbs equally at all terrestrial wavelengths. The optical thickness for this absorber is zero at the top of the atmosphere and has value τ_0 at the surface. For simplicity, assume that atmospheric pressure and density vary exponentially with constant scale height H and ignore any variation of scale height with temperature. Assume the surface is a blackbody and the atmosphere is a graybody (e.g., emits like a blackbody scaled by a fraction equal to emissivity) at terrestrial wavelengths. Also assume that optical thickness can be multiplied by a constant "diffusivity factor" r to account for radiation that goes at all angles of the hemisphere and not just in the vertical direction. For this assignment, it will be most convenient to create a model with layers that have equal thickness in pressure coordinates. Let the surface pressure be p_0.

In addition to radiative transfer, energy can be exchanged between the surface and atmosphere and within the atmosphere via turbulent transport. Assume that the transfer of sensible and latent heat between the surface and atmosphere can be represented by a bulk aerodynamic approach incorporating the difference in temperature or mixing ratio between the surface and the near-surface atmosphere, the surface wind speed V_0, and a constant exchange coefficient. Assume that transport of sensible and latent energy upward in the atmosphere occurs via mixing associated with the process of convective adjustment, which occurs when the atmosphere is unstable under either unsaturated or saturated conditions. The outcome of convective mixing can be represented as vertical averaging of conserved atmospheric properties. For simplicity, ignore virtual effects of water vapor and assume that the specific heat of the atmosphere is always that for dry air.

It will not be possible to match all characteristics of Earth's atmosphere and climate with a gray radiative transfer model and greatly simplified turbulent mixing, but we will choose parameters that approximately match the insolation and surface temperature of the tropical ocean experiencing deep convection (~303 K). First, we will assume that the atmosphere absorbs zero solar radiation and that the solar radiation absorbed by the surface is $F_{SW} = (S_0 / 4)(1 - \alpha_p)$, where S_0 is insolation and α_p is the fraction of solar radiation reflected back to space by the surface and atmosphere. A value of $S_0 / 4 = -400$ W m^{-2} is representative of the tropics, which when combined with $\alpha_p = 0.3$, results in $F_{SW} = -280$ W m^{-2} (flux defined negative downwards). This value of F_{SW} corresponds to an effective blackbody emission temperature of $T_e = 265.1$ K. We will set the total atmospheric optical thickness to a value of $\tau_0 = 1.42$ to account for greater absorption of LW radiation by more abundant water vapor, which when combined with $r = 1.66$, yields a column atmospheric absorptivity equal to 0.905. If we approximate the atmosphere as a partially transmissive single layer above a blackbody surface, then $F_{SW} = -280$ W m^{-2} combined with single-layer atmospheric emissivity $\varepsilon_a = 0.905$ produces a radiative equilibrium surface temperature $T_s = 308.2$ K and atmospheric temperature $T_a = 259.2$ K.

Finally, we will assume that the atmosphere is saturated because it is undergoing deep convection. Although saturation and convection occur within clouds, we will not model any cloud impacts on terrestrial radiation (cloud impacts on solar radiation may be incorporated into α_p). Essentially, we are assuming that the cloud area fraction is infinitesimal but the latent heating which takes place within clouds is rapidly communicated to the rest of the area via gravity waves enabled by the large Rossby radius of deformation.

1. The starting point for this assignment will be an atmosphere and water surface of depth h in radiative equilibrium. For now, we will ignore all effects of surface water on climate in the model except for thermal inertia, and we will assume that there is no turbulent exchange of sensible and latent heat between the atmosphere and water surface. However, as preparation for later questions, we will express the effect of radiative divergence as producing a change in static energy rather than a change in atmospheric temperature. Static energy is a more convenient variable for representing convection because it is conserved under adiabatic processes, unlike temperature.

   a) Provide an expression for dry static energy s in terms of height z, atmospheric temperature T_a, and any other necessary parameters. For simplicity, we will assume that geometric height and geopotential height are the same.

   b) Let ds/dt be the rate of change of dry static energy in a layer in the atmosphere. Provide an expression for ds/dt in terms of the divergence of net radiation flux $-dF/dp$ and any other necessary parameters (units J kg^{-1} s^{-1}). Note that our assumption of constant scale height H means that layers do not move up and down in elevation.

   c) Because emission of terrestrial radiation is governed by temperature rather than static energy, it is still necessary to represent temperature. Provide an expression for atmospheric temperature T_a in terms of dry static energy s and any other necessary parameters.

   d) Provide initial values of surface and atmospheric temperature to the model and iterate it forward in time by calculating how surface temperature and atmospheric dry static energy change over successive time increments. Radiative equilibrium is obtained when the net flux of solar and terrestrial radiation converges to zero at the surface and top of the atmosphere. Calculate the radiative equilibrium surface temperature $T_{s,\,rad}$ for $r = 1.66$, $\tau_0 = 1.42$, and $F_{SW} = -280$ W m^{-2}.

   e) Let $T_{a,\,rad}(z)$ be the radiative equilibrium temperature of the atmosphere in height coordinates. Plot how $T_{a,\,rad}(z)$ varies with z in the range $0 \le z \le 16$ km for $H = 8$ km, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, and $\tau_0 = 1.42$.

   f) Dry static energy is a useful metric for determining the presence of absolute instability in an atmospheric temperature profile because it decreases with height when the lapse rate exceeds the dry adiabatic lapse rate. Let $s_{rad}(z)$ be the radiative equilibrium dry static energy of the atmosphere. Plot how $s_{rad}(z)$ varies with z in the range $0 \le z \le 16$ km for $H = 8$ km, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, and $\tau_0 = 1.42$.

   g) Is the atmosphere (not considering the surface) stable or unstable under unsaturated conditions?

   h) The surface air temperature is often defined as the temperature of the air at a height 2 m above the surface. Provide an expression that extrapolates the surface air temperature $T_{2\mathrm{m}}$ from the temperature of the lowest layer in the atmosphere assuming that the layer temperature T_{a1} occurs at the layer midpoint z_{a1} and the lapse rate is Γ_d.

   i) Calculate the radiative equilibrium surface air temperature $T_{2\mathrm{m},\,rad}$ assuming that lapse rate $\Gamma_d = 0.00977$ K m^{-1}. Specify what value of $\Delta p \,/\, p_0$ you used.

j) What is the difference between radiative equilibrium surface temperature and surface air temperature $T_{s,\,rad} - T_{2\mathrm{m},\,rad}$?

k) Is the surface layer of the atmosphere in immediate communication with the surface stable or unstable?

2. In nature, the large difference between the surface temperature T_s and the surface air temperature $T_{2\mathrm{m}}$ would cause a large turbulent sensible heat flux F_{SH} that would exchange energy between the surface and atmosphere in addition to radiative transfer. This can be represented by a bulk aerodynamic formula.

   a) Provide an expression for F_{SH} in terms of T_s, $T_{2\mathrm{m}}$, surface wind speed V_0, exchange coefficient C_{SH}, and any other necessary parameters. Let turbulent sensible heat flux defined as positive upward.

   b) Calculate the sensible heat flux produced by the difference between the radiative equilibrium surface temperature and surface air temperature from Question 1(j) assuming that $V_0 = 5$ m s^{-1}, $C_{SH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

   c) The sensible heat flux is a transfer of energy between the surface and the lowest layer of the atmosphere. If we assume for now that zero energy is turbulently transferred from the lowest layer to the layer above it, provide an expression for the divergence of turbulent heat flux $-dF_H/dp$ in the lowest layer in terms of sensible heat flux F_{SH}, the pressure thickness of the layer Δp, and any other necessary parameters.

   d) Provide an expression for the rate of change of dry static energy ds/dt in the lowest layer of the atmosphere in terms of sensible heat flux F_{SH}, the pressure thickness of the layer Δp, and any other necessary parameters. Ignore other forms of energy transfer.

   e) Provide an expression for the rate of change of temperature T_{a1}/dt in the lowest layer of the atmosphere in terms of sensible heat flux F_{SH}, the pressure thickness of the layer Δp, and any other necessary parameters. Ignore other forms of energy transfer.

   f) Calculate the rate of change of temperature T_{a1}/dt in the lowest layer of the atmosphere (units K day^{-1}) produced by the sensible heat flux F_{SH} from Question 2(b). Assume all energy goes into the lowest layer and ignore other forms of energy transfer. Specify what value of Δp you used.

3. Surface sensible heat flux can be included in the model as an additional means of transferring energy between the surface and the atmosphere. Accordingly, the rate of change of surface temperature will be determined by the sum of sensible heat flux and net solar and terrestrial radiation flux. The energy transferred by sensible heat flux goes into the lowest layer of the atmosphere and is then turbulently and radiative transferred to higher layers.

   A simple implementation of turbulent transfer of energy is that convective mixing occurs between two layers whenever the higher layer has less dry static energy than the lower layer.

   $$s_{higher} < s_{lower}$$

When this condition is met, the two layers mix such that afterward they each have the same average dry static energy. The following equation applies if the two layers have equal mass.

$$s_{average} = (s_{higher} + s_{lower}) / 2$$

If dry static energy increases with height, then no turbulent transfer of energy occurs between layers.

Since the surface is much warmer than the atmosphere in the model, upward sensible heat flux will greatly increase the dry static energy of the bottom layer of the atmosphere. The bottom layer will then convectively mix with the layer above it. If the resulting average dry static energy of the bottom two layers is larger than that of the layer above them, then the bottom three layers will convectively mix. This process will continue until the bottom N layers have less average dry static energy than the N+1st layer above them. At this point, there no longer is any instability in the atmospheric profile. Dry static energy will be constant with height in the bottom N layers and increase with height in the higher layers. Temperature will decrease at the dry adiabatic lapse rate in the bottom N layers and at a smaller lapse rate in the higher layers.

Turbulent transfer of energy can be combined with radiative transfer of energy in a multistep process:

1) Calculate net radiation flux, radiative divergence, and sensible heat flux from surface and atmospheric temperatures.
2) Calculate the change in dry static energy due to radiative divergence and obtain new values of dry static energy after a time increment.
3) Add the energy of sensible heat flux accumulated over the time increment to the lowest layer of the atmosphere.
4) Starting from the bottom, convectively mix layers by averaging their dry static energy calculated in Step 2 plus the additional energy from sensible heat flux calculated in Step 3 and assign the average dry static energy to every layer in the stack.
5) Add additional layers to the convecting stack until dry static energy no longer decreases between the top of the stack and the layer above it.
6) Calculate new atmospheric temperatures from the new values of dry static energy produced by radiative divergence, sensible heat flux, and convective mixing.
7) Calculate the change in surface temperature after a time increment that is produced by the surface net solar and terrestrial radiation flux calculated in Step 2 and the sensible heat flux calculated in Step 3.
8) Return to Step 1 and repeat the cycle if the net energy flux has not yet converged to zero (below an iteration threshold) at the surface and the top of atmosphere.

a) Calculate the radiative-dry-convective equilibrium surface temperature $T_{s,\,rad\text{-}dry}$ for $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, $C_{SH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

b) Let $T_{a,\,rad\text{-}dry}(z)$ be the radiative-dry-convective equilibrium temperature of the atmosphere. On the same plot, show how $T_{a,\,rad}(z)$ and $T_{a,\,rad\text{-}dry}(z)$ vary with z in the range $0 \le z \le 16$ km for $H = 8$ km, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, $C_{SH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

c) Let $s_{rad\text{-}dry}(z)$ be the radiative-dry-convective equilibrium dry static energy of the atmosphere. On the same plot, show how $s_{rad}(z)$ and $s_{rad\text{-}dry}(z)$ vary with z in the range $0 \le z \le 16$ km for $H = 8$ km, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, $C_{SH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

d) Calculate the radiative-dry-convective equilibrium surface air temperature $T_{2\mathrm{m},\, rad\text{-}dry}$ assuming that lapse rate $\Gamma_d = 0.00977$ K m^{-1}. Specify what value of $\Delta p / p_0$ you used.

e) What is the difference between radiative-dry-convective equilibrium surface temperature and surface air temperature $T_{s,\, rad\text{-}dry} - T_{2\mathrm{m},\, rad\text{-}dry}$?

f) Calculate the sensible heat flux produced by the difference between the radiative-dry-convective equilibrium surface temperature and surface air temperature assuming that $V_0 = 5$ m s^{-1}, $C_{SH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

4. In addition to sensible heat, latent heat is turbulently transferred between the surface and atmosphere. Latent heat is a component of moist static energy, which is conserved under saturated adiabatic processes, unlike dry static energy.

a) Provide an expression for saturated moist static energy h_{sat} in terms of height z, atmospheric temperature T_a, saturation water vapor mixing ratio q_{sat}, and any other necessary parameters. For simplicity, we will assume that geometric height and geopotential height are the same.

b) A starting point for determining saturation water vapor mixing ratio is saturation vapor pressure. Provide an expression for saturation vapor pressure e_{sat} in terms of temperature T_a, a reference temperature $T_{ref} = 273$ K, a reference saturation vapor pressure $e_{ref} = 611$ Pa, and any other necessary parameters.

c) Provide an expression for saturation mixing ratio q_{sat} in terms of temperature T_a, pressure p, and any other necessary parameters.

d) Provide an expression for mixing ratio q in terms of saturation mixing ratio q_{sat} and relative humidity RH.

e) Provide an expression for the latent heat flux F_{LH} in terms of the mixing ratio right at the surface q_s, the mixing ratio 2 m above the surface $q_{2\mathrm{m}}$, surface wind speed V_0, exchange coefficient C_{LH}, and any other necessary parameters. This can be represented by a bulk aerodynamic formula. Let turbulent latent heat flux defined as positive upward.

e) Calculate the latent heat flux produced by the difference between the radiative-dry-convective equilibrium surface mixing ratio and surface air mixing ratio assuming that the surface is saturated, $\mathrm{RH}_{2\mathrm{m}} = 80\%$, $V_0 = 5$ m s^{-1}, $C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

f) Calculate the rate of change of temperature T_{a1}/dt in the lowest layer of the atmosphere (units K day^{-1}) produced by the latent heat flux F_{LH} from Question 4(e). Assume all energy goes into the lowest layer and ignore other forms of energy transfer. Specify what value of Δp you used.

g) Let $h_{sat,\ rad\text{-}dry}(z)$ be the saturated moist static energy calculated from the radiative-dry-convective equilibrium temperature of the atmosphere. Plot how $h_{sat,\ rad\text{-}dry}(z)$ varies with z in the range $0 \le z \le 16$ km for $H = 8$ km, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, $C_{SH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

h) Is the atmosphere (not considering the surface) stable or unstable under saturated conditions?

5. Surface latent heat flux can be included in the model as an additional means of transferring energy between the surface and the atmosphere. Accordingly, the rate of change of surface temperature will be determined by the sum of sensible heat flux, latent heat flux, and net solar and terrestrial radiation flux. The energy transferred by sensible and latent heat flux goes into the lowest layer of the atmosphere and is then turbulently and radiative transferred to higher layers.

The turbulent transport of latent heat can be implemented by using moist static energy rather than dry static energy in the procedure described in Question 3. Radiative divergence also changes moist static energy, and convective mixing occurs between two layers whenever the higher layer has less moist static energy than the lower layer.

$$h_{higher} < h_{lower}$$

Similarly, each has the same average moist static energy after mixing.

$$h_{average} = (h_{higher} + h_{lower}) / 2$$

If moist static energy increases with height, then no turbulent transfer of energy occurs between layers.

Since a decrease of moist static energy with height produces convective instability only under saturated conditions, we will assume that the entire atmosphere is saturated. One additional benefit of this assumption is that mixing ratio in every atmospheric layer will always be the saturation mixing ratio and therefore solely determined by temperature since the pressure of a layer does not change. This will free us from the need to track water as a separate variable and make it simple to calculate the vertical profile of moist static energy h_{sat} from the vertical profile of temperature.

However, we will need to impose two additional constraints on saturation mixing ratio to maintain physical consistency in the model. The first issue is that very warm temperatures may occur in some layers near the surface as our simple model is iterated forward. This will result in a saturation vapor pressure that is large relative to atmospheric pressure, so we must constrain saturation vapor pressure not to exceed atmospheric pressure and saturation mixing ratio not to exceed a certain value, such as 1 kg kg^{-1}. The second issue is that saturation mixing ratio, and therefore our assumed mixing ratio, will increase with height near the top of the atmosphere as pressure goes to zero. This is unphysical since the source of water vapor is at the surface, and water vapor in the stratosphere is limited by the small amount that passes through the "cold trap" at the tropopause (here we ignore oxidation of methane to water vapor in the stratosphere). Since no condensation will occur this high in the atmosphere and the stratosphere is driven by dry thermodynamics, there will be little loss of realism in the model if we constrain saturation mixing ratio to be a small constant value, such as 0.0015 kg kg^{-1}, at pressures smaller than 100 hPa.

We will continue to use the dry adiabatic lapse rate to obtain temperature at 2 m above the surface and assume that the air may have RH < 100% for the purpose of calculating latent heat flux even though these are inconsistent with our assumption that the entire atmosphere is saturated. For simplicity, we will tolerate the inconsistency rather than model an unsaturated well-mixed layer with constant mixing ratio beneath the lifting condensation level.

Finally, although it is easy to analytically calculate h_{sat} from T_a at a given z and p, it is not simple to analytically calculate T_a from h_{sat} at a given z and p. Therefore, an iterative approach can be used in which values of h_{sat} are calculated from different values of T_a until the convergence occurs on a particular value of T_a that produces the desired value of h_{sat}.

a) Calculate the radiative-saturated-convective equilibrium surface temperature $T_{s,\,rad\text{-}conv}$ for $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, RH$_{2\mathrm{m}}$ = 80%, $C_{SH} = C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}. Hereafter, radiative-saturated-convective equilibrium will be shortened to *radiative-convective* equilibrium.

b) Calculate the radiative-convective equilibrium surface air temperature $T_{2\mathrm{m},\,rad\text{-}conv}$ assuming that lapse rate $\Gamma_d = 0.00977$ K m^{-1}. Specify what value of $\Delta p / p_0$ you used.

c) What is the difference between radiative-convective equilibrium surface temperature and surface air temperature $T_{s,\,rad\text{-}conv} - T_{2\mathrm{m},\,rad\text{-}conv}$?

d) Let $T_{a,\,rad\text{-}conv}(z)$ be the radiative-convective equilibrium temperature of the atmosphere. On the same plot, show how $T_{a,\,rad}(z)$, $T_{a,\,rad\text{-}dry}(z)$, and $T_{a,\,rad\text{-}conv}(z)$ vary with z in the range $0 \le z \le 16$ km for $H = 8$ km, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, RH$_{2\mathrm{m}}$ = 80%, $C_{SH} = C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

e) Let $s_{rad\text{-}conv}(z)$ be the radiative-convective equilibrium dry static energy of the atmosphere. On the same plot, show how $s_{rad}(z)$, $s_{rad\text{-}dry}(z)$, and $s_{rad\text{-}conv}(z)$ vary with z in the range $0 \le z \le 16$ km for $H = 8$ km, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, RH$_{2\mathrm{m}}$ = 80%, $C_{SH} = C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

f) Let $h_{sat,\,rad\text{-}conv}(z)$ be the radiative-convective equilibrium saturated moist static energy of the atmosphere. On the same plot, show how $h_{sat,\,rad}(z)$, $h_{sat,\,rad\text{-}dry}(z)$, and $h_{sat,\,rad\text{-}conv}(z)$ vary with z in the range $0 \le z \le 16$ km for $H = 8$ km, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, RH$_{2\mathrm{m}}$ = 80%, $C_{SH} = C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

g) Why is the surface temperature lower for radiative-convective equilibrium than radiative-only equilibrium? Why is the atmospheric temperature higher overall for radiative-convective equilibrium than radiative-only equilibrium?

h) The troposphere in the model corresponds to the layers of the atmosphere that are convectively mixed, and the tropopause can be demarcated as the highest layer that is convectively mixed. What are the height and temperature of the tropopause for the conditions used in Question 5(f)?

6. This question will examine energetics for radiative-convective equilibrium. Provide answers assuming $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, RH$_{2\mathrm{m}}$ = 80%, $C_{SH} = 10^{-3}$, $C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

a) What is the net solar and terrestrial radiation flux at the surface?

b) What is the sensible heat flux?

c) What is the latent heat flux?

d) By what process does the surface primarily heat? By what process does the surface primarily cool?

e) Considering the atmosphere as a whole, what is the average rate of atmospheric heating caused by the latent heat of precipitation (units K day^{-1})?

f) Let $F_{LW,\ rad\text{-}conv}(z)$ be the atmospheric profile of net terrestrial radiation flux. Plot how $F_{LW,\ rad\text{-}conv}(z)$ varies with z in the range $0 \le z \le 16$ km for $H = 8$ km.

g) Let $dT_{a,\ div}/dt(z)$ be the atmospheric profile of cooling rate produced by radiative divergence. Plot how $dT_{a,\ div}/dt(z)$ varies with z in the range $0 \le z \le 16$ km for $H = 8$ km.

h) Considering the atmosphere as a whole, what is the average rate of atmospheric cooling caused by the divergence of net radiation flux (units K day^{-1})?

i) By what process does the atmosphere primarily heat? By what process does the atmosphere primarily cool?

7. This question will examine quasi-circulation for radiative-convective equilibrium. Provide answers assuming $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, $RH_{2m} = 80\%$, $C_{SH} = C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

   a) Provide an expression for vertical velocity w (change of height per unit time) assuming that diabatic loss of dry static energy due to radiative divergence $-dF/dp$ is balanced by vertical advection of dry static energy.

   b) Let $w(z)$ be the atmospheric profile of vertical velocity producing advection of dry static energy that balances the cooling rate produced by radiative divergence. Plot how $w(z)$ varies with z in the range $0 \le z \le 16$ km for $H = 8$ km.

8. This question will examine water for radiative-convective equilibrium. Provide answers assuming $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $\tau_0 = 1.42$, $V_0 = 5$ m s^{-1}, $RH_{2m} = 80\%$, $C_{SH} = 10^{-3}$, $C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

   a) Provide an expression for precipitation rate P in terms of latent heat flux F_{LH} and any other necessary parameters.

   b) Calculate the precipitation rate (units mm day^{-1})?

   c) Integrated water vapor (IWV) is the depth of liquid water that would result if all the water vapor in the atmospheric column were rained out (not including the contribution from latent heat flux). Provide an expression for IWV in terms of column average mixing ratio q_{avg} and any other necessary parameters.

   d) Calculate IWV assuming that the atmosphere is saturated (units mm).

9. We will now apply the model to investigate differences between an atmosphere with $1\times CO_2$ concentration ($\tau_0 = 1.42$) and an atmosphere with $2\times CO_2$ concentration ($\tau_0 = 1.48$). We will assume that the optical thickness of the atmosphere instantly increases from $\tau_0 = 1.42$ to $\tau_0 = 1.48$ and will first consider radiation flux before the temperatures of the surface and the atmosphere have had any time to adjust. Then we will consider temperature, energetics, quasi-circulation, and water after the surface and atmosphere have reached radiative-convective equilibrium with the new optical thickness.

   a) Calculate the terrestrial net radiation flux $F_{LW}(p)$ when the surface and atmospheric temperature correspond to radiative-convective equilibrium for $\tau_0 = 1.42$ but emissivity and transmissivity correspond to $\tau_0 = 1.48$. Let $\Delta F_{LW}(p)$ be the difference between $F_{LW}(p)$ for $\tau_0 = 1.42$ radiative-convective equilibrium and $\tau_0 = 1.42$ emissivity-transmissivity and $F_{LW}(p)$ for $\tau_0 = 1.42$ radiative-convective equilibrium and $\tau_0 = 1.48$ emissivity-transmissivity. Plot how $\Delta F_{LW}(p)$ varies with p in the range $0 \leq p \leq p_0$.

   b) The change in the net radiation flux at the top of the atmosphere after an instantaneous doubling of CO_2 is the negative of radiative forcing ΔQ. What is the value of ΔQ after τ_0 instantaneously increases from 1.42 to 1.48?

      Note that we are assuming that all of the increase in optical thickness is produced by an increase in CO_2. If we assume that a substantial part of the optical thickness increase is produced by an increase in water vapor, then only part of the change in net radiation flux would arise from a doubling of CO_2 and the true value of ΔQ would be smaller than the change in net radiation flux.

   c) Calculate the radiative-convective equilibrium surface temperature for $\tau_0 = 1.48$ and, as before, $F_{SW} = -280$ W m^{-2}, $r = 1.66$, $V_0 = 5$ m s^{-1}, $RH_{2m} = 80\%$, $C_{SH} = C_{LH} = 10^{-3}$, and $\rho = 1.2$ kg m^{-3}.

   d) The climate sensitivity parameter λ_R scales radiative forcing to the change in equilibrium surface temperature.

      $\Delta T_s = \lambda_R \, \Delta Q$

      What is the value of λ_R in the model (units K per W m^{-2})?

      Note that this value greatly underestimates λ_R for Earth's climate because the only climate feedback that it includes is a negative feedback. If we assume that a substantial part of the optical thickness increase is produced by water vapor instead of only CO_2, then ΔQ would be smaller and our calculated value of λ_R would be larger because we would be implicitly including a positive water vapor feedback.

   e) Calculate the radiative-convective equilibrium surface air temperature $T_{2m,\, rad\text{-}conv}$ assuming that lapse rate $\Gamma_d = 0.00977$ K m^{-1}. Specify what value of $\Delta p \,/\, p_0$ you used.

   f) What is the difference between radiative-convective equilibrium surface temperature and surface air temperature $T_{s,\, rad\text{-}conv} - T_{2m,\, rad\text{-}conv}$?

   g) Plot how the difference between $1\times CO_2$ concentration and $2\times CO_2$ concentration in radiative-convective equilibrium temperature of the atmosphere varies with z in the range $0 \leq z \leq 40$ km.

   h) What is the negative climate feedback that is operating in the model?

i) What are the height and temperature of the tropopause for the conditions used in Question 8(f)?

j) What is the change in radiative-convective equilibrium tropopause height between 1×CO_2 concentration and 2×CO_2 concentration? What is the change in temperature?

10. This question will examine the change in radiative-convective equilibrium energetics between 1×CO_2 concentration ($\tau_0 = 1.42$) and 2×CO_2 concentration ($\tau_0 = 1.48$).

a) What is the net solar and terrestrial radiation flux at the surface for 2×CO_2 conditions? What is the percentage change from 1×CO_2 conditions?

b) What is the sensible heat flux for 2×CO_2 conditions? What is the percentage change from 1×CO_2 conditions?

c) What is the latent heat flux for 2×CO_2 conditions? What is the percentage change from 1×CO_2 conditions?

d) Considering the atmosphere as a whole, what is the average rate of atmospheric heating caused by the latent heat of precipitation for 2×CO_2 conditions (units K day^{-1})? What is the percentage change from 1×CO_2 conditions?

e) Considering the atmosphere as a whole, what is the average rate of atmospheric cooling caused by the divergence of net radiation flux for 2×CO_2 conditions (units K day^{-1})? What is the percentage change from 1×CO_2 conditions?

11. This question will examine the change in radiative-convective equilibrium water between 1×CO_2 concentration ($\tau_0 = 1.42$) and 2×CO_2 concentration ($\tau_0 = 1.48$).

a) Calculate the precipitation rate for 2×CO_2 conditions (units mm day^{-1})? What is the percentage change from 1×CO_2 conditions?

b) Calculate IWV for 2×CO_2 conditions assuming that the atmosphere is saturated (units mm). What is the percentage change from 1×CO_2 conditions?